

National Aeronautics and Space Administration

Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) Programs

Fiscal Year 2026-2027 Broad Agency Announcement (BAA)

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Refer To Appendices for Proposal Due Dates

Disclaimer: This document is a Broad Agency Announcement Omnibus which, by itself, is not a solicitation. Solicitations will come in the form of appendices to this document that will be released as NASA needs warrant.

SBIR/STTR BAA Change Log

Version	Description of Changes	Date
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Executive Summary

The NASA SBIR/STTR Program is moving to a Broad Agency Announcement (BAA) (See more in FAR 35.016 <https://www.acquisition.gov/far/35.016>) to procure research and development activities from small businesses. This Broad Agency Announcement (BAA) sets the requirements for you, the offeror, to submit a proposal to NASA for Small Business Innovation Research (SBIR) (including SBIR Ignite) and Small Business Technology Transfer (STTR) Program Phase I and Phase II projects. NASA will solicit proposals for specific funding opportunities by releasing appendices to this BAA throughout the effective period of this BAA.

This document contains the objectives, funding information, proposal preparation instructions, and eligibility criteria. Each released appendix will contain instructions to submit a proposal package and the solicited research and technology subtopics for the specific funding opportunity. Requirements specific to each opportunity will be included in the appendices. **In any instance where specific requirements in appendices conflict with requirements in the BAA, the specific requirements in the appendices take precedence for that opportunity.** As of the release of this BAA, NASA does not have an exact number of planned appendices, but the program anticipates releasing 2 to 3 appendices each fiscal year. More information about the release schedule can be found on our website: https://www.nasa.gov/sbir_sttr/

The NASA SBIR/STTR program supports small business to create innovative, disruptive technologies that benefit society and may be used in NASA programs and missions, other government agencies, and/or sold in commercial markets. Different from most investors, the NASA SBIR/STTR Program provides equity-free funding for early or "seed" stage research and development.

The NASA SBIR/STTR program focuses on the following:

1. stimulate technological innovation
2. use small business to meet Federal R/R&D needs
3. foster and encourage participation by emerging and undercapitalized SBCs in technological innovation
4. increase private sector commercialization of innovations derived from Federal R/R&D, thereby increasing competition, productivity and economic growth

Important considerations:

Ensure you have the following registrations complete and up to date. If you are not registered, NASA recommends you start immediately.

- SAM.gov registration at <https://sam.gov/>. You must have a unique Entity Identifier (UEI)
- Registration with the SBIR Company Registry at <https://app.www.sbir.gov/company-registration/overview>

Active SAM.gov registration is required at submissions and at the time of contract award. Your North American Industry Classification System (NAICS) code must be 541713 or 541715 at the time of contract award.

Agencies must assess the security risks presented by offerors with financial ties or obligations to certain foreign countries. SBIR programs may not make awards to businesses with certain connections to foreign entities. See sections [1.1.1](#) Due Diligence Program to Assess Security Risks and [2.3.1](#) Disclosures of Foreign Affiliation or Relationships to Foreign Countries for additional details.

Important Changes for 2026-2027:

- The SBIR/STTR will be utilizing a BAA, and appendices will be released that describe the specific dates, eligibility, submissions system, and technical subtopics for that funding opportunity.
- Guidance for both Phase I and Phase II opportunities are contained in the BAA. Follow instructions carefully.
- Maximum award values have increased by 50% to \$225,000 for Phase I and \$1,275,000 for Phase II.
- TABA funds are now included within the total award value and must comply with the applicable maximum award limits.
- For SBIR and STTR Phase I proposals Part 6 has been renamed from "Market Opportunity" to "Commercialization and Business Plan."

1. Program Description

1.1 Legislative Authority and Background

Congress created the Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) programs to support scientific excellence and technological innovation through the investment of federal research funds. The purpose of this investment is to build a strong national economy, strengthen the role of small business in meeting federal research and development needs, increase the commercial application of research results, and foster and encourage participation by socially and economically disadvantaged and women-owned small businesses.

The [Small Business Administration \(SBA\)](#) provides policy through the combined Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) programs. The [Small Business Innovation and Economic Security Act](#) amended the Small Business Act (*15 U.S.C. 638*) to extend the SBIR and STTR programs until September 30, 2031. The NASA SBIR/STTR Programs follow the policies and practices of the [SBA SBIR/STTR Policy Directive](#). Revisions to the Policy Directive can be viewed through the [Federal Register Notice](#).

1.1.1 Due Diligence Program to Assess Security Risks

Title 15 of the United States Code 638, section vv, [\[15 U.S.C. 638 \(vv\)\]](#), requires NASA, in coordination with the SBA, to establish and implement a due diligence program to assess security risks presented by offerors seeking a federally funded award. As noted above, this revision is incorporated into this solicitation, including Appendix III, “Disclosures of Foreign Affiliations or Relationships to Foreign Countries” as reflected in the Disclosures of Foreign Affiliations or Relationships to Foreign Countries form (see section [2.3.1](#)).

In accordance with [15 U.S.C 638 \(vv\)](#), NASA will review all proposals submitted in response to this solicitation to assess security risks presented by offerors seeking a SBIR or STTR award. NASA will use information provided by the offeror in response to the Disclosures of Foreign Affiliations or Relationships to Foreign Countries form and the proposal to conduct a risk-based due diligence review on the cybersecurity practices, patent analysis, employee analysis, and any foreign ownership of a small business concern, including the financial ties and obligations (which shall include surety, equity, and debt obligations) of the offeror and its employees to a foreign country, foreign person, or foreign entity. Additionally, the due diligence review will conduct a risk-based due diligence review of foreign affiliations of a covered individual, owner, or other key personnel of a small business concern with an entity in a foreign country of concern; investment relationships of small business concern with an individual or entity in a foreign country of concern; technology licensing agreements or joint ventures (including joint venture like agreements) with an individual or entity in a foreign country of concern; and ‘business relationships between a covered individual, owner, or other key personnel of a small business concern and an individual or entity in a foreign country of concern. This review includes any foreign risk connecting the small business concern to an entity, including any affiliates of the entity, or individual on any relevant prohibited lists defined by law. NASA will also assess proposals utilizing open-source analysis and analytical tools, all valid means for the nondisclosures of the information set forth in 15 U.S.C. § 638 (9)(g)(17).

1.2 Purpose and Priorities

This BAA sets the requirements for you, the offeror, to submit a proposal to NASA for Small Business Innovation Research (SBIR)/Small Business Technology Transfer (STTR) Programs. This BAA includes both Phase I and Phase II opportunities. Only SBCs that are eligible to submit a Phase II proposal will be allowed to submit. Unless otherwise noted in the appendices, you must follow the proposal requirements set in this BAA. Furthermore, program background information, eligibility requirements for participants, information on the three program phases, and information for preparing proposal packages are contained herein. **NASA will release appendices containing funding opportunity details throughout the effective period of the BAA.**

The NASA SBIR/STTR Program Management Office (PMO), hosted at the NASA Ames Research Center, operates the programs together with NASA mission directorates, programs, projects, and centers. The NASA Shared Services Center (NSSC) manages SBIR and STTR procurements.

Each year NASA mission directorates, programs, projects, and centers identify the research problems and technology needs that the SBIR/STTR program will solicit. The range of problems and technologies is broad, and the list of research subtopics varies from year to year to maintain alignment with current NASA priorities.

The purpose of Phase II is to continue the R/R&D developed under Phase I with a goal to commercialize and bring the innovation to either Federal or commercial markets, or both, as the Phase II project completes. Phase II will require a more comprehensive application, outlining the proposed effort in detail and the commercialization strategy for the effort. When further research is required to bring the project results to the marketplace, completion of the research under these circumstances may have to be carried into Post Phase II opportunities or Phase III.

A separate BAA will not be issued requesting Phase II proposals, and unsolicited proposals will not be accepted. All proposing SBCs that receive a Phase I award from this BAA will be eligible to participate in Phase II competitions and potential Phase III awards.

1.3 Three-Phase Program

NASA SBIR/STTR projects advance through three phases and are described in detail on the NASA SBIR/STTR website: https://www.nasa.gov/sbir_sttr/.

Phase I

Phase I projects should demonstrate technical feasibility of the proposed innovation and the potential for use in a NASA program or mission and/or the commercial market. The NASA SBIR/STTR Program does not make awards solely directed toward system studies, market research, routine engineering, development of existing product(s), proven concepts, or modifications of existing products without substantive innovation.

Maximum value and period of performance (POP) for Phase I (unless otherwise noted in the appendix for the specific opportunity):

Phase I Contracts	SBIR	SBIR Ignite	STTR
Maximum Contract Value	\$225,000	\$225,000	\$225,000
Period of Performance	6 months	6 months	13 months

Phase II

Phase II proposals continue the research and development started in Phase I to bring the innovation closer to use in a NASA program or mission and/or the commercial market. Phase II requires a more detailed proposal of the technical effort and commercialization strategy. Only Phase I awardees are eligible to submit a Phase II proposal. Maximum value and period of performance (POP) for Phase II (unless otherwise noted in the appendix for the specific opportunity):

Phase II Contracts	SBIR	SBIR Ignite	STTR
Maximum Contract Value	\$1,275,000	\$1,275,000	\$1,275,000
Maximum Period of Performance	24 months	24 months or less	24 months

Post-Phase II Opportunities for Continued Technology Development

Phase I and II awards may not be sufficient in either dollars or time to prepare the project for government or commercial use. Therefore, NASA supports small businesses beyond Phase I and II awards with several Post Phase II initiatives. Please refer to the [NASA SBIR/STTR website](#) for eligibility, application deadlines, matching requirements and further information.

Phase III

SBIR/STTR awardees are eligible to receive sole-source Phase III contracts any time after award of their Phase I contracts. In Phase III, customers outside the SBIR and STTR programs—including NASA programs, other government agencies, or the private sector—fund the further development or use of innovative technologies, products, and services resulting from either a Phase I or Phase II award. Please refer to the NASA SBIR/STTR website for [Phase III information](#).

1.4 Availability of Funds

NASA does not commit to fund any proposal or to make a specific number of awards. NASA may elect to make several or no awards in any specific research subtopic. NASA will determine the number of awards based on the level of appropriated funding provided to the program.

1.5 Eligibility Requirements

1.5.1 Small Business Concern (SBC)

You must submit a certification stating that the SBC meets the size, ownership, and other requirements of the SBIR program at the time of proposal package submission, award, and at any other time set forth in SBA's regulations at [13 CFR §§ 121.701-121.705](#).

1.5.2 SBC Size

You, combined with affiliates, must not have more than 500 employees.

1.5.3 Restrictions on Level of Small Business Participation

You must be the primary performer of the proposed research effort. SBIR and STTR opportunities have different requirements for the percentage of work that must be performed by you and any subcontractors, consultants, or research institutions.

SBIR (including SBIR Ignite)

Phase I: To be awarded an SBIR Phase I contract, you must perform at least two-thirds or 67% of the effort, and subcontractors or consultants may perform up to one-third or 33% of the effort. See section [3.1.5.4](#).

Phase II: To be awarded an SBIR Phase II contract, you must perform a minimum of 50% of the research or analytical effort during Phase II; correspondingly, an outside party such as consultants or subcontractors may perform a maximum of 50% of the effort. Involvement of university, government, or other outside personnel in the planning and research stages of the project as consultants or through subcontracting arrangements is permitted and may be particularly helpful to small businesses.

STTR

Phase I and II: To be awarded a STTR contract, you must perform at least 40% of the effort, and a single research institution must perform at least 30% of the effort.

1.5.4 Place of Performance and American-Made Products and Equipment

Congress intends that the awardee of a Funding Agreement under the SBIR/STTR program should, when purchasing any equipment or a product with funds provided through the Funding Agreement, purchase only American-made equipment and products, to the extent possible, in keeping with the overall purposes of this program.

If a rare and unique circumstance exists (for example, if a supply, material, equipment, product, subcontractor/consultant, or project requirement is not available in the United States), NASA requires you to provide justification by completing the [Foreign Vendor Form](#). This form must be submitted with your Proposal Budget documentation, see section [3.1.5.4](#). NASA will consider a deviation request during contract negotiation and either approve or decline before award.

If a foreign vendor is proposed, the Phase I contract may be delayed or not awarded. NASA will not approve purchases from or work with countries that appear on the Designated Country list. For reference, please see <https://www.nasa.gov/oirr/export-control>.

1.5.5 Principal Investigator (PI) Employment Requirement

Phase I and II PI Requirements	SBIR (including Ignite)	STTR
Primary Employment:	PI must be primarily employed with the SBC.	PI must be primarily employed with the SBC or Research Institution (RI).
Employment Certification: Primary employment means that more than one-half of the PI employment time is spent in the employ of the SBC or RI, based on a 40-hour workweek.	The PI must be primarily employed with the SBC at the time of award and during the conduct of the proposed project.	The PI must be primary employed with the SBC or the RI at the time of award and during the conduct of the proposed project.
Co-PIs	Not allowed	Not allowed
Deviation Request	NASA will review any deviation requests during negotiation and either approve or decline before award.	NASA will review any deviation requests during negotiation and either approve or decline before award.
Misrepresentation of Qualifications	If you misrepresent qualifications, NASA will decline the proposal package or terminate the contract.	If you misrepresent qualifications, NASA will decline the proposal package or terminate the contract.
Substitution of PIs	To substitute PIs, you must request approval from NASA after award	To substitute PIs, you must request approval from NASA after award

Note: NASA considers full-time employment to include salaried employees and employees who regularly work a 40-hour workweek. NASA considers a 19.9-hour or more workweek elsewhere to conflict with this rule. In rare occasions, minor deviations from this requirement may be necessary; however, any minor deviation must be approved in writing prior to the award by the Contracting Officer after consultation with the NASA SBIR/STTR Program Manager.

1.5.6 Novated/Successor in Interested/Revised Funding Agreements

An SBIR or STTR awardee may include, and work may be performed by, those identified via a “novated” or “successor in interest” or similarly revised funding agreement. In addition, an awardee may include those that have merely reorganized with the same key staff, regardless of whether they have been assigned a different tax identification number. In cases where there is a novation or similarly revised funding agreement, agencies may require the original awardee to relinquish its rights and interests in an SBIR project in favor of another applicant as a condition for that applicant's eligibility to participate in the programs for that project.

1.5.7 Restrictions on Venture-Capital-Owned Businesses

Unless otherwise noted, small businesses owned in majority part by multiple venture capital operating companies (VCOs), hedge funds, or private equity firms are not eligible to submit a proposal to this solicitation. See detailed definitions at [29 CFR 2510.3-101\(d\)](#)

1.5.8 Joint Ventures or Limited Partnerships

Both joint ventures and limited partnerships are permitted, provided the entity created qualifies as an SBC as defined in [1.5.1](#). Include in the proposal package a copy or summary of the joint venture or partnership agreement that

includes, at a minimum, a statement of how the workload will be distributed, managed, and charged. See definitions for Joint Ventures along with examples at [13 CFR 121.103\(h\)](#).

1.5.9 Required Benchmark Transition Rate

More experienced firms (SBCs with 21 or more Phase I awards) must meet performance benchmark requirements to continue participating in SBIR and STTR programs. The purpose of these benchmarks is to ensure that Phase I offerors that have won multiple prior SBIR and STTR awards are progressing towards commercialization. SBA will notify companies failing the benchmarks as well as the relevant officials at participating agencies like NASA.

Please refer to <https://www.sbir.gov/performance-benchmarks> for more information.

1.6 NASA Technology Available (TAV) for SBIR Use

You may use technology developed by NASA, or Technology Available (TAV), on SBIR projects. NASA has over 1,400 patents available for licensing, including many patents related to sensors and materials, and over 1,000 available software applications/tools in the Portfolio and Software Catalog via the NASA Technology Transfer Portal, <http://technology.nasa.gov>.

NASA provides these technologies "as is" and makes no representation or guarantee that additional effort will result in infusion or commercial viability. Whether or not an offeror proposes the use of a NASA patent or computer software within its proposed effort will not in any way be a factor in the selection for award.

1.6.1 Use of NASA Software

If you intend to use NASA software, a Software Usage Agreement (SUA), on a nonexclusive, royalty-free basis, is necessary, and the clause at [48 C.F.R. 1852.227-88](#), Government-Furnished Computer Software and Related Technical Data, will apply to the contract. Awardees will request the SUA from the appropriate NASA Center Software Release Authority (SRA) after contract award.

1.6.2 Use of NASA Patent

If you intend to use a NASA patent, you must apply for a nonexclusive, royalty-free evaluation license prior to submitting a proposal. After you have identified a patent to license in the NASA patent portfolio (<http://technology.nasa.gov>), click the link on the patent webpage ("Apply to License") to NASA's Automated Licensing System (ATLAS) to finalize your license with the appropriate field center technology transfer office. You must provide the completed evaluation license application with the proposal following the directions in section [3.1.5.8](#).

Inventor Knowledge Transfer

An SBIR/STTR awardee that has been granted a nonexclusive, royalty-free evaluation license to use a NASA patent under the SBIR/STTR award may, if available and on a noninterference basis, also have access to NASA personnel knowledgeable about the NASA patent. Licensing executives located at the appropriate NASA field center will be available to assist awardees with this request; however, access to the inventor for the purpose of knowledge transfer will require the requestor to enter into a non-disclosure agreement (NDA) or other agreement, such as a Space Act Agreement. The awardee may also be required to reimburse NASA for knowledge transfer activities. **Preparation of the agreements for knowledge transfer may require a significant amount of time to complete; therefore, NASA does not recommend pursuing inventor access for Phase I projects.**

1.7 I-Corps™(Phase I Only)

NASA partners with the [National Science Foundation \(NSF\)](#) to give **Phase I** awardees the opportunity to participate in the [NSF Innovation Corps \(I-Corps™\)](#) program. I-Corps requires you to conduct customer discovery to learn your customers' needs, to obtain a better understanding of your company's value proposition, and to develop an outline of

a business plan for moving forward. This training is designed to lower the market risk inherent in bringing a product or innovation to market, thereby improving the chances for a viable business. For more information on the NASA I-Corps program, visit the [NASA SBIR/STTR website](#).

If you are selected for Phase I contract negotiations, you will be provided the opportunity to opt into and participate in the NASA SBIR/STTR I-Corps program as indicated in section [3.1.5.10](#).

For SBIR and SBIR Ignite Phase I awardees the amount of funding is up to \$18,000 and for STTR Phase I awardees the amount of funding is up to \$30,000 to support participation in the I-Corps Program. I-Corps awards will be made separately with a modification with the Phase I contract.

1.8 Technical and Business Assistance (TABA)

Under the [Small Business Act](#), you may request Technical and Business Assistance (TABA) for **Phase I (up to \$6,500)** and **Phase II (up to \$50,000)**. TABA requests will be treated as part of the overall award amount and must remain within the established maximum award thresholds.

If your project is selected for award and the TABA is approved, you must use the TABA to contract with one or more vendors to assist in:

- Making better technical decisions on SBIR/STTR projects;
- Solving technical problems that arise during SBIR/STTR projects;
- Minimizing technical risks associated with SBIR/STTR projects; and/or
- Commercializing the SBIR/STTR product or process, including intellectual property protections.

For additional approved and restricted uses of TABA funding, see https://www.nasa.gov/sbir_sttr/taba/.

TABA vendors may include private commercialization assistance or business development service providers, public-private partnerships, other entrepreneurial support organizations (ESOs), and attorneys or other IP or licensing professionals. NASA does not guarantee approval of requests for TABA.

For information on requesting TABA, please see section [3.1.5.9](#). Additional TABA information can also be found in sections [3.1.5.4](#), [4.7](#), [5.3](#), and [5.4](#).

1.9 Small Business Administration (SBA) Applicant Resources

The SBA works with several local partners of various organizational types to train and support potential SBIR/STTR applicants around the country from proposal assistance to SAM registration, and commercialization support to industry connections. To find local assistance visit: <https://www.sbir.gov/local-assistance>.

To find out more information on the specific types of SBA federal resources available, visit: <https://www.sbir.gov/resources>.

1.10 NASA Mentor-Protégé Program (MPP)

The purpose of the NASA Mentor-Protégé Program (MPP) is to provide incentives to NASA contractors, performing under at least one active approved subcontracting plan negotiated with NASA, to assist protégés in enhancing their capabilities to satisfy NASA and other contract and subcontract requirements. The NASA MPP under the authority of [48 CFR Parts 1819 and 1852](#) and managed by the Office of Small Business Programs (OSBP), includes an Award Fee Pilot Program. Under the Award Fee Pilot Program, a mentor is eligible to receive an award fee at the end of the agreement period based upon the mentor's performance of providing developmental assistance to an active SBIR/STTR Phase II contractor in a NASA Mentor-Protégé Agreement (MPA). For more information on the Mentor-Protégé Program, please visit <https://www.nasa.gov/osbp/mentor-protége-program/>.

1.11 Fraud, Waste and Abuse and False Statements

Fraud is “any false representation about a material fact or any intentional deception designed to deprive the United States unlawfully of something of value or to secure from the United States a benefit, privilege, allowance, or consideration to which an individual or business is not entitled.”

NASA reserves the right to decline any proposal packages that include plagiarism and false claims. Further, knowingly and willfully making any false, fictitious, or fraudulent statements or representations may be a felony under the Federal Criminal False Statement Act ([18 U.S.C., section 1001](#)), punishable by a fine and imprisonment of up to 5 years in prison. The [Office of the Inspector General](#) (OIG) has full access to all proposal packages submitted to NASA.

Pursuant to NASA policy, any company representative who observes crime, fraud, waste, abuse, or mismanagement or receives an allegation of crime, fraud, waste, abuse, or mismanagement from a federal employee, contractor, grantee, contractor, grantee employee, or any other source will report such observation or allegation to the OIG. NASA contractor employees and other individuals are also encouraged to report crime, fraud, waste, and mismanagement in NASA's programs to the OIG. The OIG offers several ways to report a complaint:

NASA OIG Hotline: 1-800-424-9183 (TDD: 1-800-535-8134)

NASA OIG Cyber Hotline: <https://oig.nasa.gov/hotline/>

Or by mail:

NASA Office of Inspector General
P.O. Box 23089
L'Enfant Plaza Station
Washington, DC 20026

1.12 NASA Procurement Ombudsman Program

The [NASA Procurement Ombudsman Program](#) is available under this solicitation as a procedure for addressing concerns and disagreements concerning the terms of the solicitation, the processes used for evaluation of proposal packages, or any other aspect of the SBIR procurement. The clause at NASA Federal Acquisition Regulation (FAR) Supplement (NFS) [1852.215-84](#) (“Ombudsman”) is incorporated into this solicitation.

The cognizant ombudsman is:

Geoffrey Sage, Procurement Ombudsman
Office of Procurement
NASA Headquarters
Washington, DC 20546-0001
Email: nhq-dl-op-comp-advocate-vendor-engagement@mail.nasa.gov

The ombudsman does not participate in any way with the evaluation of proposal packages, the source selection process, or the adjudication of formal contract disputes. Therefore, before consulting with the ombudsman, you must first address your concerns, issues, disagreements, and/or recommendations to the Contracting Officer for resolution. The process set forth in this solicitation provision does not change your right to file a bid protest or the period in which to timely file a protest.

1.13 General Information

1.13.1 Questions About This Solicitation and Means of Contacting NASA SBIR/STTR Program

To ensure fairness, NASA will not answer questions about the intent and/or content of research subtopics in funding opportunity during the open submissions period unless part of a public question and answer opportunity.

If you have questions requesting clarification of proposal package instructions and administrative matters, refer to the NASA SBIR/STTR website or contact the NASA SBIR/STTR Help Desk.

Reference the funding opportunity appendix for Help Desk guaranteed timely answer deadline.

1. NASA SBIR/STTR Website: https://www.nasa.gov/sbir_sttr/
2. Help Desk:
 - a. Email: agency-sbir@mail.nasa.gov
 - b. You must provide the name and telephone number of the person to contact, the organization name and address, and the specific questions or requests.

1.13.2 Fiscal Year vs Program Year

The NASA SBIR/STTR program uses fiscal year (FY) and program year (PY) to track our funding opportunities and awards. Fiscal year refers to the Federal fiscal year and is used by the program to denote which fiscal year funds are being utilized. Program year refers to the year the funding opportunity was made available. Phase II awards are tracked by the program year the Phase I offering was made. For example, PY2026 Phase II opportunities will be available in this FY26-27 BAA to firms that received PY2026 Phase I awards.

1.14 Definitions

NASA strongly encourages you to review the list of definitions available at https://www.nasa.gov/sbir_sttr/program-definitions/. These definitions include those from the combined SBIR/STTR Policy Directives as well as terms specific to NASA.

2. Registrations, Certifications and Other Information

2.1 Small Business Administration (SBA) Company Registry

You must register with SBA's Company Registry and update your commercialization status. See <https://app.www.sbir.gov/company-registration>. You must provide your unique SBC Control ID (assigned by SBA upon completion of the Company Registry registration) and upload a PDF copy of the SBA Company Registry registration with the Firm Certification Form.

2.2 System for Award Management (SAM) Registration

SAM, maintained by the GSA's Federal Acquisition Service, is the primary repository for contractor information required to conduct business with NASA. To be registered in SAM, all mandatory information, including the Unique Entity Identifier (UEI) and a Commercial and Government Entity (CAGE) code, must be validated in SAM. You may obtain information on SAM registration and annual confirmation requirements at <https://sam.gov/content/home> or by calling 866-606-8220.

Per FAR 4.1102(a), you must be registered in SAM:

- At the time of proposal submission
- At the time of contract award
- During contract performance.

For more information, review the Federal Register (89 FR 89472 and 90 FR 38206).

To be eligible for SBIR awards, you must have an active SAM registration under North American Industry Classification System (NAICS) code 541713 or 541715 as a small business at the time of award.

If you are not registered, apply for registration immediately upon receipt of this solicitation. Typically, SAM registration and updates to SAM registration take several weeks. In order to receive an SBIR/STTR award from NASA, purpose of registration must be listed as "All Awards" on your SAM Registration.

Note that your SAM registration Doing Business As (DBA) name will appear on all contract documents.

2.3 Certifications

You must complete the Firm and Proposal Certifications by answering "Yes" or "No" to certifications as applicable in the electronic submissions system. Carefully read each of the certification statements. The Federal Government relies on the information to determine whether you are eligible for a SBIR program award.

NASA uses a similar certification to ensure continued compliance with specific program requirements at time of award and at the time of final payment. The definitions for the terms used in this certification are set forth in the Small Business Act, SBA regulations (13 CFR Part 121), the SBIR/STTR Policy Directives, and any statutory and regulatory provisions referenced in those authorities.

For **Phase I** awards, in addition to invoice certifications and as a condition for payment, a life cycle certification must be completed in Contracts ProSAMS (<https://my.prosams.nasa.gov/>). The life cycle certification must be completed along with the final invoice certification before uploading the final invoice in the Department of Treasury's Invoice Processing Platform (IPP).

For **Phase II** awards, two life cycle certifications must be completed in Contracts ProSAMS. A life cycle certification must be completed along with the second invoice certification as a condition of payment of the second invoice. Another life cycle certification must be completed along with the final invoice certification as a condition of payment of the final invoice. The life cycle certifications are preset in the ProSAMS.

If the Contracting Officer believes that you may not meet certain eligibility requirements for award, they may request you provide clarification or additional supporting documentation. If the Contracting Officer still believes you are not eligible, you must file a protest with the SBA, who will determine eligibility.

2.3.1 Disclosures of Foreign Affiliation or Relationships to Foreign Countries

You must complete the “Disclosures of Foreign Affiliations or Relationships to Foreign Countries” form as part of your proposal submission. Even if you do not have any foreign relationships, you must complete this form to represent that such relationships do not exist. If you do not submit this form, NASA will decline your proposal during the administrative screening process, and it will not be evaluated. Foreign involvement or investment does not independently disqualify your proposal, but failing to disclose such affiliations or relationships may result in denial of an award.

The disclosures require the following information:

- (A) the identity of all owners and covered individuals of the small business concern who are a party to any foreign talent recruitment program of any foreign country of concern, including the People’s Republic of China;
- (B) the existence of any joint venture or subsidiary of the small business concern that is based in, funded by, or has a foreign affiliation with any foreign country of concern, including the People’s Republic of China;
- (C) any current or pending contractual or financial obligation or other agreement specific to a business arrangement, or joint venture-like arrangement with an enterprise owned by a foreign state or any foreign entity;
- (D) whether the small business concern is wholly owned in the People’s Republic of China or another foreign country of concern;
- (E) the percentage, if any, of venture capital or institutional investment by an entity that has a general partner or individual holding a leadership role in such entity who has a foreign affiliation with any foreign country of concern, including the People’s Republic of China;
- (F) any technology licensing or intellectual property sales to a foreign country of concern, including the People’s Republic of China, during the five-year period preceding submission of the proposal; and
- (G) any foreign entity, offshore entity, or entity outside the United States related to the small business concern;
- (H) the identity and relationship of all owners, officers, and covered individuals that have foreign affiliations with a research institution located in a foreign country of concern.

After reviewing the above listed disclosures, and if determined appropriate by NASA, the program may ask you clarifying questions related to your disclosures and/or to provide true copies of any contractual or financial obligation or other agreement specific to a business arrangement or joint venture-like arrangement with an enterprise owned by a foreign state or foreign entity in effect during the five-year period before proposal submission.

For the purposes of the above disclosures, a “covered individual” means an individual who contributes in a substantive, meaningful way to the scientific development or execution of a research and development project proposed to be carried out or has access to, or could reasonably be considered to have access to, the technology development being proposed.

During award, updated disclosures must be provided within 30 days of any substantive changes to the project, to include any changes to a required disclosure.

2.4 Federal Acquisition Regulation (FAR) and NASA Certifications and Clauses

SAM contains required certifications that you may access at <https://www.acquisition.gov/browsefar> as part of the required registration (see FAR 4.1102). You must complete these certifications to be eligible for award. You must provide representations and certifications electronically via the website and update the representations and certifications as necessary, and at least annually, to keep them current, accurate, and complete. NASA will not enter any contract if you do not comply with these requirements.

In addition, you will need to be aware of the clauses that will be included in the contract if selected for a contract. For a complete list of FAR and NASA clauses see the funding opportunity appendix.

2.5 Software Development Standards

If you are proposing projects involving the development of software, you may be required to comply with NASA Procedural Requirements (NPR) 7150.2D, NASA Software Engineering Requirements, available online at https://nodis3.gsfc.nasa.gov/npg_img/N_PR_7150_002D/N_PR_7150_002D_Preface.pdf.

2.6 Human and/or Animal Subject

NASA requires a protocol approved by a NASA review board if proposed work includes human or animal subjects. **Due to the complexity of the approval process, NASA does not allow use of human and/or animal subjects for Phase I projects.** Reference 14 CFR 1230 and 1232.

2.7 Flight Safety Standards

If you are proposing projects involving the delivery of a spacecraft, you must comply with NASA Procedural Requirements (NPR) 8079.1, NASA Spacecraft Conjunction Analysis and Collision Avoidance for Space Environment Protection, available online at <https://nodis3.gsfc.nasa.gov/displayDir.cfm?t=NPR&c=8079&s=1>.

2.8 Commercial Air Service and NASA Airworthiness Requirements

All flight activities are subject to NASA STD 7919.1 - Commercial Air Service (CAS) Standard; NPR 7900.3D – Aircraft Operations Management; and Center-specific requirements, such as AFOP-7900.3-023 – Airworthiness & Flight Safety Review Process.

These review requirements may preclude SBIR/STTR Phase I flight activities due to their short contractual period, combined with ongoing NASA flight project reviews and staffing availability. As a result, flight activities planned for SBIR/STTR Phase II efforts are much more realistic.

Any desired flight elements or flight activities under the SBIR/STTR program are encouraged to work with NASA Armstrong Flight Research Center (AFRC) SBIR/STTR staff to initiate flight planning activities aboard NASA AFRC support aircraft. Additionally, integrating into the existing NASA AFRC flight planning and review process for flight on NASA AFRC aircraft expedites these requirements.

2.9 Homeland Security Presidential Directive 12 (HSPD-12)

If your project is selected for award and requires access to federally controlled facilities or access to a federal information system (as defined in [FAR 2.101](#)) for 6 consecutive months or more, you must apply for and receive appropriate Personal Identify Verification (PIV) credentials.

FAR [52.204-9](#), Personal Identity Verification of Contractor Personnel, states in part that the contractor must ensure that individuals needing such access provide the personal background and biographical information requested by NASA. See <https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.201-3.pdf>.

3. Proposal Preparation Instructions and Requirements

3.1 Requirements to Submit a Proposal Package

3.1.1 General Requirements

NASA will be using electronic submissions tools to accept proposal packages. This BAA guides firms through the steps for submitting a complete proposal package. All submissions will be completed through the secure electronic submissions tool listed in the funding opportunity specific appendix. Most communication between NASA and the firm is through email.

Guidelines for preparing both Phase I and Phase II proposals are provided below, but you are only eligible to submit a Phase II proposal during your Phase I period of performance. To be eligible to receive a Phase I award, you must submit a complete Phase I proposal package. You may only submit one Phase II proposal per Phase I award.

A complete proposal package must contain all required content as described in section [3.1.5](#) below.

Be thoughtful in selecting a subtopic to ensure the proposal is responsive to the topic. **NASA will not move a proposal between subtopics, programs or funding opportunities.**

3.1.1.1 Classified Information

NASA will decline any proposal package that contains classified information.

3.1.2 Phase I

Proposal packages contain a Technical Proposal as described in section [3.1.5.5](#) below. A Phase I Technical Proposal must clearly and concisely:

1. Describe the proposed innovation relative to the current state of the art;
2. Address the scientific, technical, and commercial merit and feasibility of the proposed innovation as well as its relevance and significance to NASA interests as described in chapter 9 of this solicitation; and
3. Provide a preliminary strategy that addresses key technical, market, and business factors pertinent to the successful development and demonstration of the proposed innovation and its transition into products and services for NASA missions and/or programs, commercial markets, and other potential markets and customers.

3.1.3 Phase II

The Phase II proposal is to continue the R/R&D developed under Phase I with a goal to commercialize and bring the innovation to Federal and commercial markets as the Phase II project completes. Phase II will require a more comprehensive application, outlining the proposed effort in detail and the commercialization strategy for the effort. NASA recognizes that Phase II awards may not be sufficient in either dollars or time for the firm to complete the total R/R&D and the commercialization activities required to bring the project results to the marketplace. Therefore, completion of the research under these circumstances may have to be carried into NASA Post-Phase II or Phase III opportunities.

Phase II proposals are more comprehensive than those required for Phase I. Submission of a Phase II proposal is in accordance with Phase I contract requirements and is voluntary. NASA assumes no responsibility for any proposal preparation expenses.

The proposal will provide the planning for a focused project that builds upon Phase I results and encompasses technical, market, financial, and business factors relating to the development and demonstration of the proposed innovation and its transition into products and services for NASA mission programs, other Government agencies (Federal and State), and non-Government markets.

3.1.4 Format Requirements

NASA administratively screens all elements of a proposal package for compliance with format requirements. At its discretion, NASA may decline any proposal package or disregard specific proposal content that exceeds the stated limits when adjusted to comply with format requirements.

Required Page Limits

A technical proposal—all parts including all graphics—must not exceed the page limit requirements of that opportunity as listed in the table below:

	SBIR		SBIR Ignite		STTR	
	Phase I	Phase II	Phase I	Phase II	Phase I	Phase II
Maximum Page Limit	15	46	15	30	15	46

NASA will not accept technical proposal uploads with any page(s) over the required page limits. The additional forms required for proposal package submission do not count against the page limits.

Suggested Page Limits for Technical Proposal Parts

Suggestions for page limits for each of the technical proposal parts can be found in the BAA Reference Guide document located in the [Firms Library](#) on our website.

Margins

Use 1.0-inch (2.5 cm) margins on all sides.

Type Size

Use type size 10 point or larger for text or tables, except as legends on reduced drawings.

Header/Footer Requirements

You may include the SBC name, proposal number, and project title in the header on each page of the proposal. Include the page number and proprietary legend (see section [3.4](#)), if applicable in the footer on each page of the proposal. You may use margins for header/footer information.

Project Title

The proposal project title must be concise and descriptive of the proposed effort. Do not use the NASA research subtopic title, acronyms, or words like "development of" or "study of."

Templates

Templates for the technical proposal and other forms may be available in the [Firms Library](#). Use of the templates is not required but is highly recommended to ensure compliance with formatting requirements.

3.1.5 Proposal Package

A specific funding opportunity may require different information. This list provides opportunity specific notes in parentheses after each item if it is not required by all opportunities. Each proposal package must contain the following items:

1. Proposal Contact Information
2. Proposal Certifications, electronically endorsed
3. Proposal Summary (must not contain proprietary data)
4. Proposal Budget
 - a. Letters of commitment for Government resources and subcontractors/consultants (if applicable)

- b. Foreign Vendor form (if applicable) – Note: NASA and the Office of Management and Budget (OMB) have issued a policy that requires a review of any request to purchase materials or supplies from foreign vendors.
- c. Subcontractor or RI budget (if applicable)
- d. Pro Forma Financial Projections Worksheet (**SBIR Ignite Phase II only**)
- 5. Technical Proposal
- 6. Briefing Chart (must not contain proprietary data)
- 7. NASA Evaluation License Application, only if NASA Technology Available (TAV) is being proposed
- 8. Capital Commitments Addendum (optional) (**SBIR/STTR Phase II, SBIR Ignite Phase I and II**)
- 9. Technical and Business Assistance (TABAs) request (optional)
- 10. I-Corps Interest Form (Phase I opportunities only)**
- 11. SBC-Level Forms
 - a. Firm Information
 - b. Firm Certifications
 - c. Audit Information
 - d. Disclosures of Foreign Affiliations or Relationships to Foreign Countries Audit Information
 - e. Prior Awards Addendum
 - f. Commercial Metrics Report (CMR)
- 12. STTR Research Agreement, allocation of rights, and endorsement of agreement by the Research Institution (RI) official (section [3.1.5.13](#)) (**STTR only**)
- 13. Electronic Endorsement by the designated small business representative and principal investigator (PI) is completed before the deadline

For many of the required forms, you can view sample forms located in the NASA SBIR/STTR Resources website: https://www.nasa.gov/sbir_sttr/firms_library/.

Note: Letters expressing general technical interest are not required or desired and will not be considered during the review process. However, if submitted, such letter(s) will count against the page limit. Letters of funding support commitments are allowable for Phase II proposals. Letters of funding support commitments should be submitted as part of the Capital Commitments Addendum as described under Part 6: Commercialization and Business Plan.

What Not to Include

NASA will not consider the following items during evaluation:

- Letters of interest or support
- Technical papers
- Product samples
- Videos
- Slides
- PowerPoint slide decks
- Other ancillary items

However, all submitted content other than the required forms designated in 1-14 above will count against the proposal page limit.

3.1.5.1 Proposal Contact Information Form

You must provide complete information for each contact person and submit the form as required. **Contact Information is public information and may be disclosed.**

3.1.5.2 Proposal Certifications Form

You must provide complete information for each question in the form and certify its accuracy as required.

3.1.5.3 Proposal Summary Form

You must provide complete information for each section of the form as required. **The Proposal Summary, including the technical abstract, is public information and may be disclosed.**

3.1.5.4 Proposal Budget Form

You must complete the Proposal Budget form following the instructions provided in the appendix. NASA will use the submitted budget information to determine whether the proposed price is fair and reasonable. See [5.5](#) Profit for Fee and [5.6](#) Cost Sharing.

The total requested funding for the proposed effort must not exceed the dollar amounts in the table below (unless otherwise noted in the opportunity-specific appendix):

Maximum Budget Request	SBIR		SBIR Ignite		STTR	
	Phase I	Phase II	Phase I	Phase II	Phase I	Phase II
Maximum TABA request amount (OPTIONAL)	\$6,500	\$50,000	\$6,500	\$50,000	\$6,500	\$50,000
Maximum total budget (with or without TABA)	\$225,000	\$1,275,000	\$225,000	\$1,275,000	\$225,000	\$1,275,000

Proposal Preparation Costs

NASA is not responsible for any monies you expend for proposal preparation and submission.

TABA

If requesting funds for TABA, your request must be included in your proposal and budget. See section [1.8](#) and [3.1.5.9](#) for more information on the TABA opportunity.

PI

Note that if the Principal Investigator is working less hours than other proposed direct labor elements or if there are individuals listed as direct labor that are not employees of your firm, it shall be explained in your proposal.

Documentation

Provide documentation, such as a quote, previous purchase order, published price lists, etc. for all proposed costs. Failure to submit this documentation could lead to delays in the processing/negotiation of your contract and could ultimately result in proposal selection being withdrawn by NASA and no contract awarded to your firm.

Additional Budget Information

In addition, submit the following information in the Proposal Budget form, as applicable:

- **Justification for submitted rates. Submit one of the following in order to justify the submitted rates for your proposal:**
 - 1. Approved rate agreement or provisional rate agreement with DCAA
 - 2. Mathematical cost basis of estimate on how rates were developed
(You can use your own template, or you can utilize the DCAA ICE Template found at <https://www.dcaa.mil/Checklists-Tools/ICE-Model/>)
- **Use of a Foreign Vendor.** If you are requesting to purchase products and equipment from a foreign vendor, you must complete the Foreign Vendor Form (see section [1.5.4](#) for more information).
- **Use of Government Resources.** If you plan to use government resources (such as, services, equipment, facilities, laboratories, etc.), as described in Part 7 of the technical proposal instructions, you must provide the following:

1. Statement, signed by the appropriate Federal department or agency official, verifying that the resources are available during the proposed period of performance, authorizing their use, and if applicable, including the associated cost.
2. Signed letter on your company letterhead explaining why your SBIR or STTR research project requires the use of government resources. Include data that verifies the absence of non-federal facilities or personnel capable of supporting the research effort, and, if applicable, the associated cost estimate.

Due to the complexity and length of time for the approval process, NASA strongly discourages you from requesting the use of government resources during the performance of a Phase I. Approval for the use of government resources for a Phase I technical proposal requires a strong justification at the time of submission and will require approval by the Contracting Officer during negotiations if selected for award.

If successful in receiving an award, you are responsible for reimbursing NASA for resources utilized in the performance of the effort, and the cost of such resources will be included in your contract values (not to exceed capped amounts).

- **Use of Subcontractors and Consultants.** You may establish business arrangements with other entities or individuals to perform some of the proposed R/R&D effort, within the limits in section [1.5.3](#) and below. Subcontractors' and consultants' work must also be performed in the United States (see section [1.5.4](#) for more information).

If you propose using subcontractors or consultants, submit the following with your proposal:

1. List of consultants by name with the number of hours and hourly costs identified for each consultant.
2. Subcontractor (to include universities) budget that aligns with your Proposal Budget form and includes direct labor, other direct costs, and profit, as well as indirect rate agreements.
3. A letter of commitment for each subcontractor and/or consultant, dated and signed by the appropriate person with contact information.
 - a. If a university is proposed as a subcontractor, the signed letter must be on the university letterhead from the Office of Sponsored Programs.
 - b. If an independent consultant is proposed, the signed letter must not be on university letterhead.

The proposed subcontracted business arrangements, including consultants, must not exceed the limits described in Section [1.5.3](#) Restrictions on Level of Small Business Participation. To calculate this percentage, divide the total cost of the proposed subcontracting effort including applicable indirect rates such as overhead and G&A by the total price proposed less profit. For examples, see documentation in the BAA Reference Guide located in the https://www.nasa.gov/sbir_sttr/firms_library/.

Percentage of subcontracting effort = (Subcontractor cost + G&A) / (Total price – Profit)

RI Budget (STTR only)

The RI's budget must be submitted at the time of proposal submission, and if the RI is an educational institution, the RI must submit a letter from the institution's Office of Sponsored Programs.

Deviations from subcontractor performance-of-work requirements

For STTR: Deviations from these STTR requirements are not allowed, as the performance-of-work requirements are specified in statute at FAR NFS [1852.219-82](#).

For SBIR and SBIR Ignite: Occasionally, deviations from this requirement may occur, and must be approved in writing by the Contracting Officer after consultation with the NASA SBIR PMO.

See Part 8 of the Technical Proposal for additional information on the use of subcontractors and consultants.

Travel in Phase I

Due to the intent and short period of performance of the Phase I contracts, along with a limited budget, NASA strongly discourages travel during the Phase I contract. If the purpose of the meeting cannot be accomplished via videoconference or teleconference, you must justify the trip in the Proposal Budget form. The Contracting Officer and Technical Monitor will review travel requests to determine if they are necessary to complete the proposed effort.

Pro Forma Financial Projections Template (only Ignite Phase II)

Ignite Phase II: Complete the “Pro Forma Financial Projections” in your Phase II Proposal Budget Form. Provide realistic financial projections based on your best estimates and market research. Avoid overly optimistic or inflated figures.

Milestone Plan (only Phase II)

For Phase II, you must submit a proposed quarterly milestone plan with the Proposal Budget form. The milestone plan must be in accordance with the proposed work plan, outlining the work to be accomplished each quarter and the cost proposed associated with each of the quarterly milestones. The cost breakdown must be similar to the Proposal Budget form for each of the proposed quarterly milestones (i.e., each milestone should include the labor, supplies, travel, and profit associated with those tasks to be accomplished that quarter). The proposed cost associated with each quarterly milestone must be realistic for the work to be accomplished but is not required to be equally distributed across each quarter.

3.1.5.5 Technical Proposal

The technical proposal must contain all 9 parts in order, number, and title as listed below. NASA will decline any proposal package that does not have all 9 parts, and it will not be evaluated. If a part is not applicable to your proposed effort, you must include the part and mark it “Not applicable.” Do not include any budget data in the technical proposal.

Read the requirements for each part carefully as requirements differ for Phase I and II proposals given the expected advancement in the proposed technology and the refinement of the commercialization plan during any Phase I efforts.

The technical proposal must provide all information described in the nine parts below. Evaluators will not seek additional information. You should clearly explain the significance of your proposed technology in terms that are understandable by someone that is not a technical expert in your field. **Review the funding opportunity specific rubric in the attachments of the appropriate appendix to ensure your proposal aligns to the evaluation criteria.**

Table of Contents

The technical proposal may begin with a brief table of contents indicating the page numbers of each of the parts.

Part Title	Page #
Table of Contents	
Part 1: Identification and Significance of the Innovation	
Part 2: Technical Objectives	

Part 3: Work Plan	
Part 4: Related R/R&D	
Part 5: Key Personnel	
Part 6: Commercialization and Business Plan	
Part 7: Facilities and Equipment	
Part 8: Subcontractors/Consultants	
Part 9: Related, Essentially Equivalent, and Duplicate Proposals and Awards	

Part 1: Identification and Significance of the Proposed Innovation

Succinctly describe:

- The proposed innovation.
- The relevance and significance of the proposed innovation to an interest, need, or needs, within a subtopic described in chapter 9 of the specific funding opportunity appendix.
- The proposed innovation relative to the current state of the art.

Phase II only: You should include relevant results from your Phase I effort in the Phase II proposal.

Part 2: Technical Objectives

State the specific objectives of the Phase I/II R/R&D effort as it relates to the problem statement(s) posed in the topic description and the types of innovations being requested. Indicate the proposed deliverables at the end of the Phase I/II effort and how these align with the proposed topic found in [section 9](#) of the specific funding opportunity appendix. Address any technical risks and associated mitigation plans. Use data to support your claims and provide references as appropriate.

If you plan to use NASA TAV including Intellectual Property (IP), you must describe planned developments with the IP. Add the NASA Evaluation License Application as an attachment in the Proposal Certifications form (see [section 1.6](#)).

Part 3: Work Plan

Include a detailed plan to meet the proposed technical objectives. The plan must include:

- Detailed task descriptions, that is, what will be done, where it will be done, and the methods you will use to do it
- Schedules
- Resource allocations
- Estimated task hours for each key personnel that match hours reported in the Proposal Budget Form
- Planned accomplishments (including project milestones)
- If the offeror is a joint venture or limited partnership, a statement of how the workload will be distributed, managed, and charged

STTR only: The work plan will specifically address the percentage and type of work to be performed by the SBC and the RI. The plan will provide evidence that the SBC will exercise management direction and control of the performance of the STTR effort, including situations in which the PI may be an employee of the RI.

Part 4: Related R/R&D

Describe significant existing R/R&D that is directly related to the technical proposal including any conducted by the PI or by the company. Describe how it relates to the proposed effort and any planned coordination with outside sources. You must demonstrate awareness of key recent R/R&D conducted by others in the specific subject area. Include any pertinent references or publications.

Part 5: Key Personnel and Bibliography of Directly Related Work

Phase I and II: Identify all technical and business personnel involved in proposed activities whose expertise and functions are essential to the success of the project and future commercialization efforts. Provide biographical information, including directly related education and experience. Where the resume/vitae are extensive, you may summarize the most relevant experience or publications. If any team ability gaps exist, provide any plans to address those gaps if possible. Use the application prompts related to “Team” in the funding opportunity specific rubrics to ensure you have included all necessary information.

The PI is key to the success of the effort. The following applies:

- **Functions:** The PI plans and directs the project, leading it technically and making substantial personal contributions during its implementation. The PI also serves as the primary contact with NASA on the project and ensures that work proceeds according to contract agreements. Competent management of PI functions is essential to project success. You must describe the nature of the PI's activities and the amount of time that the PI will personally apply to the project. The amount of time the PI proposes to spend on the project must be acceptable to the Contracting Officer.
- **Qualifications:** You must clearly present the qualifications and capabilities of the proposed PI and the basis for PI selection. NASA has the sole right to accept or decline a PI based on factors such as education, experience, demonstrated ability and competence, and any other evidence related to the specific assignment.
- **Eligibility:** You must establish and confirm the eligibility of the PI and indicate if existing projects and other proposals recently submitted or planned commit the time of the PI concurrently with this proposed project. NASA will decline your proposal if you try to circumvent the restriction on PIs working more than half time for an academic or a nonprofit organization by substituting an ineligible PI.

Phase II: If the Phase II PI or key personnel are different than the key personnel under Phase I, please provide rationale for the change.

Part 6: Commercialization and Business Plan

Describe your potential commercialization approach or commercialization and business plan. Use the guidance below and the application prompts and evaluation criteria found in the funding opportunity specific rubrics in the appendix to develop the content for this Part. Any technical discussion in Part 6 should be understandable to someone without technical expertise in your specific technology.

The SBIR/STTR program is mandated to move funded innovations into federal and private sector commercial markets. Companies that address market opportunities early are better positioned to apply for and receive follow-on SBIR/STTR contracts and to commercialize their innovations. NASA encourages you to use TABA (Phase I and II) and I-Corps (Phase I only), to help you address market opportunities. See sections [3.1.5.9](#) for how to request TABA and [3.1.5.10](#) for opting into I-Corps.

SBIR and STTR Phase I: Describe the potential commercialization approach for the innovation by addressing the following:

- The potential economic benefits associated with your innovation.
- The potential customers and basic go-to-market strategy.
- The potential risks in bringing your innovation to market.

SBIR and STTR Phase II: NASA requires Phase II offerors to provide commercialization and business planning information in the proposal and is enforcing a requirement that firms provide a minimum amount of information as required by page length. Firms that do not meet this requirement may be considered nonresponsive to this requirement and the proposal may not receive a score for this section which could result in an overall lower score for the proposal.

The Commercialization Plan should provide the following information to communicate and validate that the firm has the knowledge and ability to commercialize the innovation being proposed.

1. Market Feasibility and Competition Strategy
2. Commercialization Strategy
3. Financial Strategy
4. Intellectual Property
5. Assistance and Mentoring

Note: Letters of funding support commitments should be submitted as part of the Capital Commitments Addendum found in section [3.1.5.7](#). Any formal letters of commitment or intent submitted in the Capital Commitments Addendum will not count toward the page limits of the proposal.

SBIR Ignite Phase I and II: Because SBIR Ignite is commercialization focused, NASA seeks to fund firms that will be successful in obtaining external investment beyond government sources. To strengthen your Phase I and Phase II proposals, you should provide evidence of follow-on funding commitments. Examples include:

1. A letter of commitment for follow-on funding and/or product sales.
2. A letter of commitment for matching funding to be provided for a future Phase II-E application.
3. A letter of capital commitment, signed by the proper authority (CEO, CFO, etc.), that indicates a commitment to provide funding and/or product sales, should the Phase II project be successful, and the market need still exists.
4. Letter of intent to provide funding should the Phase II project be successful, and the market need still exists.
5. A specific plan to secure Phase III funding.

Note: The proposal should only include a list of these letters/commitments. The actual letters should be combined in a single document that is uploaded separately in the “Capital Commitments Addendum” upload link in the electronic submissions system.

Part 7: Facilities/Equipment

Describe the types, location, and availability of equipment necessary to carry out the work proposed. You must justify any proposed equipment purchase. **When purchasing equipment or a product under the SBIR/STTR contract, you should purchase only American-made products or equipment.**

Phase I: Although use of government-furnished laboratory equipment, facilities, or services (collectively, “government resources”) is strongly discouraged in Phase I proposals, describe in this part why the use of such government resources is necessary and not reasonably available from the private sector if government resources are required. See sections [3.1.5.4](#) and [5.13](#) for additional requirements when proposing use of such government resources. The narrative description of resources should support the proposed approach and documentation in the Proposal Budget form.

If you plan to use a federal laboratory/facility during a follow-on Phase II contract, please state this intent in your Phase I proposal.

Phase II: If you request to use Government-furnished laboratory equipment, facilities, or services (collectively, “Government resources”), describe in this part why the use of such Government resources is necessary and not reasonably available from the private sector. See sections [3.1.5.4](#) and [5.13](#) for additional requirements when proposing use of such Government resources. The narrative description of resources should support the proposed approach and documentation in the Budget form.

Part 8: Subcontractors and Consultants

Describe all subcontracting or other business arrangements, including who they are with and for what expertise, functions, services, and number of hours. You must ensure that all organizations and individuals are available for the time periods proposed. The narrative description of subcontractors and consultants in the technical proposal should support the proposed approach and documentation in the Proposal Budget form, section [3.1.5.4](#). **If partnering is not required to successfully mature your technology, please explain why.**

Part 9: Related, Essentially Equivalent, and Duplicate Proposals and Awards

WARNING: It is illegal to enter into multiple funding agreements for essentially equivalent work. While you may submit similar or identical proposals to multiple solicitations, it is risky. You must notify the agencies in advance and resolve the matter prior to award.

If you choose to submit identical proposals or proposals containing a significant amount of essentially equivalent work under other federal program solicitations, you must include a statement in each proposal containing:

1. The name and address of the agencies to which proposals were submitted or from which awards were received.
2. Date of proposal submission or date of award.
3. Title, number, and date of solicitations under which proposals were submitted or awards received.
4. The specific applicable research subtopics for each proposal submitted or award received.
5. Titles of research projects.
6. Name and title of principal investigator or project manager for each proposal submitted or award received.

A summary of essentially equivalent work information, as well as related research and development on proposals and awards, is also required on the Proposal Certifications form (if applicable).

3.1.5.6 Briefing Chart

The briefing chart information is required and will be used by evaluators to assist in the ranking of technical proposals prior to selection. Summarize the following on the provided electronic form:

- Identification and Significance of Innovation
- Technical Objectives and Proposed Deliverables
- NASA Applications
- Non-NASA Applications
- Graphic

The briefing chart is public information and may be disclosed. Do not include proprietary information or International Traffic in Arms Regulation (ITAR)-restricted data. For more information on ITAR see <https://www.sbir.gov/tutorials/itar/>.

3.1.5.7 Capital Commitments Addendum

SBIR and STTR Phase II and SBIR Ignite Phase I and II:

The Capital Commitments Addendum applies to SBIR Phase II, STTR Phase II, and SBIR Ignite Phase I and II proposals. Letters of capital commitment act as an indication of market validation for the proposed innovation and add significant credibility to the proposed effort. Although NASA can be a future stakeholder/buyer and could

possibly issue a Phase III contract for the innovation to be infused in a future mission or program, NASA's goal is for small businesses to commercialize innovations in markets that are larger than just NASA. Your Phase II or Ignite Phase I letters of capital commitment should demonstrate that the company has initiated dialogue with relevant stakeholders in commercial markets that are relevant to NASA (potential customers, end users, strategic partners, investors, etc.) for the proposed innovation and that a legitimate business opportunity may exist should the innovation prove feasible outside of NASA.

The letter(s) must not exceed 2 pages in length; should come directly from potential customers, end users, strategic partners, investors, etc.; and must contain affiliation information and contact information for the signatory stakeholder. Letters and supporting documents that only support the development of the innovation with no capital funding commitment as described above will not be reviewed. Letters and supporting documents from state, local, and Congressional representatives are NOT considered letters of capital commitment. They should not be submitted as part of the application and will not be reviewed.

If letters of capital commitment are not appropriate for this stage of innovation due to business considerations, then the offeror must clearly justify why letters of capital commitment are not being included in the proposal. The justification should relate to the technical and commercial considerations of the innovation proposed in the application.

3.1.5.8 NASA Evaluation License Application, only if TAV is being proposed

If you applied for TAV by following the instructions found at <http://technology.nasa.gov>, upload the application with your proposal package.

3.1.5.9 Request for Technical and Business Assistance (TABA)

You may request TABA funds (see section [1.8](#)) for **Phase I (up to \$6,500)** and **Phase II (up to \$50,000)** as part of your contract value within the maximum award limits. If you request TABA funds, you must do so in the proposal package submission. You are not required to request TABA at Phase I or Phase II. Eligibility for requesting TABA funds at Phase II is not dependent on Phase I TABA participation. NASA cannot direct you to any specific TABA vendor or website. Therefore, you will choose your own TABA vendor(s).

TABA requests must be made within the Proposal Budget form and the electronic submissions system, where you will also submit your quote(s). You must provide a quote(s) that includes the following from each vendor:

1. Contact information of the vendor (name, address, phone number, email address, and website)
2. Description of vendor(s) expertise and knowledge of providing the desired TABA services
3. Itemized list of services. This should include:
 - a. Description of tasks associated with each service and expected outcome
 - b. Cost for each service. This includes the number of hours and hourly cost for each service, when appropriate.
4. Description of the deliverable(s) the TABA vendor will provide for each service. This is generally in the form of a report(s).

All TABA vendors must be legal businesses in the United States. NASA will consider TABA requests that are missing any requested TABA information as incomplete and will not review the TABA request or provide TABA approval under the award.

The TABA is treated as part of your Phase I and Phase II contract award value and is not subject to any profit or fee by the requesting offeror and cannot be used in the calculation of indirect cost rates or general and administrative expenses (G&A). The TABA cost(s) and service(s) to be provided by each vendor will be

based on the period of performance. NASA will not consider requests for TABA funding outside of your proposal package submission or contractual period of performance.

NASA reserves the right to withhold funds requested for TABA until a formal review and approval of the requested vendor(s) is completed. If the project is selected for the award and you demonstrate this requirement sufficiently as determined by NASA, NASA will permit you to acquire the requested assistance as an allowable cost of the SBIR/STTR award. Approval or denial of TABA funding will be provided during the contract negotiation period and before the award.

NASA encourages you to use the limited amount of \$6,500 Phase I TABA funds for:

1. Development of a Phase II TABA Needs Assessment – If you plan to request TABA at Phase II, you should secure a TABA vendor at Phase I to support the development of a Phase II TABA needs assessment. The goal of the TABA Needs Assessment is to determine and define the types of TABA services and costs you would need if the project was selected for a future Phase II award. Phase II TABA may be up to \$50,000.
2. Development of a Phase II Commercialization and Business Plan – If you are planning to submit a future proposal for Phase II funding, you will be required to submit a commercialization and business plan that meets the requirements of that future Phase II solicitation. NASA encourages you to use Phase I TABA to secure a TABA vendor to help develop the commercialization and business plan. The goal of the commercialization and business plan is to allow NASA to evaluate your ability to commercialize the innovation and provide a level of confidence regarding your future and financial viability.

3.1.5.10 I-Corps Interest Form (Phase I only)

You will complete a short I-Corps interest form as part of your proposal package submission. NASA uses this form to determine the level of interest from Phase I offerors to participate in the NASA I-Corps program. See section [1.7](#).

Based on the initial level of interest in the I-Corps program, NASA plans to open the opportunity to all Phase I awardees to ensure a successful cohort of teams participate in the program. Phase I awardees will receive a Call for I-Corps Proposal from the SBIR/STTR PMO during contract negotiations describing the process to provide a 5-page proposal to participate in the I-Corps program. NASA will provide directions for completing the proposal including due dates, training dates, and available funding by email. NASA reserves the right to limit the number of offerors to participate in the I-Corps program based on the assessment of the I-Corps proposals and funding availability.

3.1.5.11 SBC Level Forms

You must complete all SBC level information within the electronic submissions system. Any required uploads do not count toward the page limit for the technical proposal.

A. Firm Information

You must complete the SBC identifying information. The information provided will be applied to all proposals you submit to this solicitation.

B. Firm Certifications

You must complete the Firm Certifications section by answering “Yes” or “No” as applicable. An example of the certifications can be found in the NASA SBIR/STTR Resources website:
https://www.nasa.gov/sbir_sttr/firms_library.

C. Audit Information

Although you are not required to have an approved accounting system, it is easier for NASA to determine that your rates are fair and reasonable if you have an approved accounting system. To assist NASA, you must complete the questions in the Audit Information form regarding your rates and upload the Federal

agency audit report or related information that is available from the last audit. There is a separate Audit Information section in the Proposal Budget form that you must also complete. If you have never been audited by a federal agency, then answer "No" to the first question, and you do not need to complete the remainder of the form. An electronic form will be provided during the submissions process. The Contracting Officer uses this Audit Information to assist with negotiations if the proposal package is selected for award. The Contracting Officer will advise you what is required to determine reasonable cost and/or rates in the event the Audit Information is not adequate. Please refer to section [3.1.5.4](#) in regard to what information can be submitted to justify proposed rates.

D. Disclosures of Foreign Affiliations or Relationships to Foreign Countries

Each offeror is required to complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries form as required in the electronic submissions system. See section [2.3.1](#) for additional information on these disclosures. You must answer "Yes" or "No" as applicable and provide the requested information related to each "yes" response.

Please note that even if you do not have any foreign relationships, you must complete the "Disclosures of Foreign Affiliations or Relationships to Foreign Countries form" to represent that such relationships do not exist. Failure to complete and include this form will result in the declination of your application during the administrative screening.

E. Prior Awards Addendum

If you have received more than 15 Phase II awards in the prior 5 fiscal years, submit the name of the awarding agency, solicitation year, phase, date of award, funding agreement/contract number, and subtopic title for each Phase II. If you have received any SBIR or STTR Phase II awards, even if fewer than 15 in the last 5 years, NASA still recommends that you complete this form as the information will be useful to you when completing the Commercialization Metrics Report (CMR).

A template is available in the [Firms Library](#).

F. Commercialization Metrics Report (CMR)

NASA uses a commercialization report/data-gathering process to track the overall commercialization success of its SBIR and STTR programs. You must complete the Commercialization Metrics Report or update an existing report if applicable, via <https://www.sbir.gov> (the report is available in the "My Dashboard" section of your company's sbir.gov profile) as part of the proposal package submissions process. Companies with no SBIR/STTR awards or awards within the last 3 to 5 years will not be penalized under past performance for the lack of past SBIR/STTR commercialization.

If you have received any Phase III awards resulting from work on any NASA SBIR or STTR awards, provide the related Phase I or Phase II contract number, name of Phase III awarding agency, date of award, Funding Agreement number, amount, project title, and period of performance. The report will also ask you to provide financial, sales, and ownership information, as well as any commercialization success you have had because of SBIR or STTR awards. You must update this information annually during proposal package submission.

CMR input is kept confidential and will not be made public except in broad aggregate, with no company-specific attribution. Do not submit password protected documents.

3.1.5.12 Contractor Responsibility Information (Phase II only)

No later than 10 business days after the notification of selection for negotiations, you must provide a signed statement from the firm's financial institution(s), on the financial institution's letterhead, stating whether or not the firm is in good standing and how long the firm has been with the institution.

3.1.5.13 Research Agreement (STTR)

For STTR proposals or when RIs are subcontractors, the Research Agreement (different from the Allocation of Rights Agreement; see https://www.nasa.gov/sbir_sttr/program-definitions/) is a single-page document electronically submitted and endorsed by the SBC and RI. A model agreement is provided, or firms can create their own custom agreement. The Research Agreement must be submitted as required in chapter 6.

All STTR Phase proposals must provide sufficient information to convince NASA that the proposed SBC/RI cooperative effort represents a sound approach for converting technical information residing at the RI into a product or service that meets a need described in a solicitation research topic.

3.1.5.14 Allocation of Rights Agreement (STTR)

To receive a STTR award, you must negotiate a written agreement between the SBC and the partnering RI, IP rights, if any, to carry out follow-on research, development, or commercialization. This written agreement must be signed by authorized representatives of the SBC, RI, and subcontractors and consultants, as applicable.

The SBC must submit this agreement with the proposal. This will help to expedite contract negotiations.

3.2 Multiple Proposal Submissions

Refer to the specific appendix for limitations on the number of proposals you may submit.

3.3 Understanding the Patent Landscape

You should indicate in the proposal that a comprehensive patent review has been completed to ensure that there is no existing patent or perceived patent infringement based on the innovation proposed. The U.S. Patent and Trade Office (USPTO) has an online patent search tool that can be found at <https://www.uspto.gov/patents-application-process/search-patents>.

3.4 Proprietary Information in the Proposal Submission

Limit proprietary information to only that information that is essential to your proposal.

Information contained in unsuccessful proposals remains your property. The Federal Government may, however, retain copies of all proposals. Public release of information in any proposal submitted will be subject to existing statutory and regulatory requirements. If proprietary information is provided in a proposal, which constitutes a trade secret, commercial or financial information, it will be treated in confidence, to the extent permitted by law, provided that the proposal is clearly marked as follows:

- (A) The following “italicized” legend must appear on the title page of the proposal:

This proposal contains information that shall not be disclosed outside the Federal Government and shall not be duplicated, used, or disclosed in whole or in part for any purpose other than evaluation of this proposal, unless authorized by law. The Government shall have the right to duplicate, use, or disclose the data to the extent provided in the resulting contract if award is made as a result of the submission of this proposal. The information subject to these restrictions is contained on all pages of the proposal except for pages [insert page numbers or other identification of pages that contain no restricted information]. (End of Legend); and

- (B) The following legend must appear on each page of the proposal that contains information you wish to protect:

Use or disclosure of information contained on this sheet is subject to the restriction on the title page of this proposal.

3.5 Release of Certain Proposal Information

In submitting a proposal, you agree to permit the government to disclose publicly the information contained in the Contact Information form and Proposal Summary form, which includes the Technical Abstract and Briefing Chart. Other proposal data is your property, and NASA will protect it from public disclosure to the extent permitted by law, including requests submitted under the Freedom of Information Act (FOIA).

4. Method of Selection and Evaluation Criteria

The NASA SBIR/STTR Program does not make awards solely directed toward system studies, market research, routine engineering, development of existing product(s), proven concepts, or modifications of existing products without substantive innovation.

4.1 Selection Process and Evaluation Criteria

NASA conducts a multi-stage review process of all proposal packages:

1. Administrative review for compliance with Chapters 3 and 6 of the solicitation
2. Initial screening for responsiveness to the subtopic
3. Technical and commercial evaluation on a competitive basis (as an “other competitive procedure” in accordance with FAR 6.102(d)(2) and FAR 35.016), using the criteria and procedures set forth within this solicitation and accompanying appendices.
4. Price evaluation
5. Scoring and weighting to determine rating
6. Prioritization
7. Selection
8. Determination of cost/price reasonableness and responsibility

For Phase II proposals, the commercial review will be completed by a panel of 3rd party commercialization reviewers who are experts in determining commercial potential.

Do not assume that evaluators are acquainted with your company, key individuals, or with any experiments or other information. NASA will judge each proposal on its own merit and will not conduct any tradeoff analyses between or among competed proposals.

4.1.1 Administrative Review

NASA will review all proposal packages received by the published deadline to determine if the proposal package meets the requirements found in chapters 3 and 6 (released in the opportunity-specific appendix). NASA may decline and not evaluate a proposal package that is not compliant with the requirements in chapters 3 and 6. NASA will notify you of its decision to eliminate the proposal package from consideration and the reason(s) for the decision.

4.1.2 Technical Responsiveness

NASA will screen proposal packages that pass the administrative review to determine technical responsiveness to the subtopic of this solicitation. Proposal packages that are not responsive to the subtopic will be declined and not evaluated. NASA will notify you of its decision to eliminate the proposal package from consideration and the reason(s) for the decision. **Ensure your technical proposal is responsive to the subtopic. NASA will NOT evaluate a technical proposal under a subtopic other than the one you select. Failure to adequately communicate how the proposal relates to the technical need posed in the subtopic will result in the proposal being rejected and no further evaluations will occur.**

4.1.3 Commercial Potential Evaluation Criteria

NASA will evaluate proposal packages that comply with administrative requirements and are technically responsive to the subtopic of the solicitation. Subject matter experts will determine the most promising technologies from a commercial potential perspective based on the criteria described in the commercial potential evaluation rubric.

For details about these criteria, please refer to the appropriate appendix where the evaluation criteria, weighting, and definitions are provided. Please review the rubric to ensure your submission addresses all areas being evaluated.

4.1.4 Technical Evaluation Criteria

NASA will evaluate proposal packages that comply with administrative requirements and are technically responsive to the topic in the opportunity-specific appendix. Subject matter experts will determine the most promising technical and scientific approaches based on the criteria listed in the rubric for the opportunity.

For more details about these criteria, please refer to appropriate appendix where the evaluation criteria, weighting, and definitions are provided. Please review the rubric to ensure your submission addresses all areas being evaluated.

4.1.5 Price Evaluation

Utilizing the procedures set forth in [FAR 15.404-1](#), NASA will evaluate the budget proposal form to determine whether the proposed pricing is fair and reasonable. NASA will only make an award when the price is fair and reasonable and approved by the NASA Contracting Officer.

If a proposal is selected for award, the Contracting Officer will review all the evaluations for the proposal and will address any pricing issues identified during negotiation of the final award.

4.2 Scoring and Weighting to Determine the Most Highly Rated Proposals

NASA will score each criterion numerically as detailed in the opportunity-specific rubric. Numerical scores are then weighted per the opportunity-specific rubric and summed to reach the proposal rating. The most highly rated proposals are eligible for prioritization.

4.3 Prioritization

For the most highly rated proposals, NASA will prioritize those proposals that offer the best solutions to the technical needs as defined in the subtopics to make recommendations to the Source Selection Official (SSO). NASA may consider a variety of additional programmatic balance factors such as portfolio balance across NASA programs, centers and mission directorates, available funding, first-time awardees/participants, and/or geographic distribution when making recommendations.

4.4 Selection

The SSO makes the final decisions to determine the proposals that will enter contract negotiations. The SSO may consider the additional programmatic balance factors identified in section [4.3](#) along with the technical merit and commercial potential.

After the SSO selection has been finalized, NASA will post the list of proposals selected for negotiation on the NASA SBIR/STTR website. All SBCs selected by the SSO will receive a formal notification letter. NASA will evaluate each proposal selected for negotiation for cost/price reasonableness. After completion of evaluation for cost/price reasonableness and a determination of responsibility, the Contracting Officer will negotiate and award an appropriate contract to be signed by both parties before work begins.

4.5 Determination of cost/price reasonableness and responsibility

Each proposal package selected for negotiation by the SSO will be evaluated by the Contracting Officer to determine eligibility for an award. The terms and conditions of the contract will be negotiated based on the SBIR Small Business Act (15 U.S.C. 638), FAR and NASA FAR requirements, and a responsibility determination will be made. The Contracting Officer will advise the SSO on matters pertaining to price analysis and responsibility determinations. A Contracting Officer will negotiate an appropriate contract to be signed by both parties before work begins.

4.6 I-Corps Evaluation Process (Phase I only)

For awardees that submit an I-Corps proposal pursuant to sections [1.7](#) and [3.1.5.10](#), NASA will provide a programmatic assessment based on the following criteria:

- Proposed team members demonstrate a commitment to the requirements of the I-Corps program.
- The proposed team includes the proper composition and roles as described in the I-Corps proposal requirements.
- The I-Corps proposal demonstrates that there is potential for commercialization in both NASA and commercial markets.

Based on the assessment of the above criteria, the NASA SBIR/STTR PMO will provide a recommendation to the SSO of I-Corps proposals to receive funding. The SSO will make the final selections.

4.7 TABA Evaluation

NASA conducts a separate review of all Phase I and II requests for TABA after the SSO makes the final selection of proposals to enter negotiation for a Phase I and Phase II contract. The SBIR/STTR PMO conducts the evaluation of the TABA request to determine if the request meets the requirements found in sections [1.8](#) and [3.1.5.9](#) and informs the Contracting Officer of the final determination to allow TABA funding under your contract. NASA will notify you of the approval or denial of your TABA request prior to award.

4.8 Access to Proprietary Data by Non-NASA Personnel

4.8.1 Non-NASA Reviewers

In addition to utilizing government personnel in the review process, NASA, at its discretion and in accordance with [1815.207-71](#) of the NASA FAR Supplement, may utilize individuals from outside the government with highly specialized expertise not found in the government. Qualified experts outside of NASA (including industry, academia, and other government agencies) may assist in performing evaluations as required to determine or verify the merit of a proposal package. In deciding to obtain an outside evaluation, NASA will take into consideration requirements for the avoidance of organizational or personal conflicts of interest and any competitive relationship between the prospective contractor or subcontractor(s) and the prospective outside evaluator. Outside evaluators will certify that the information (data) contained in the proposal package is for evaluation purposes and will not be further disclosed.

4.8.2 Non-NASA Access to Confidential Business Information

In the conduct of proposal package processing and potential contract administration, NASA may need to provide access to the proposal package to other NASA contractor and subcontractor personnel. NASA will provide access to such data only under contracts that contain an appropriate NFS [1852.237-72](#) Access to Sensitive Information clause that requires the contractors to fully protect the information from unauthorized use or disclosure.

4.9 Notification and Feedback to Offerors

After selections for negotiation have been made, NASA will send a notification to the designated small business representative identified in the proposal package according to the processes described below.

Due to the competitive nature of the program and limited funding, recommendations to fund or not fund a proposal package are final. NASA will not reconsider selection decisions or provide additional feedback or information regarding the final decision. Offerors are encouraged to use the written feedback to understand the outcome and review of their proposal package and to develop plans to strengthen future proposals.

Unsuccessful Phase II offerors cannot resubmit their unsuccessful Phase II proposal to a future Phase II solicitation and if planning to apply for a new Phase I, the Phase I proposal must be different and cannot duplicate the prior Phase I work.

4.9.1 Feedback

NASA uses a two-stage process to notify offerors of the outcome of their proposal package.

1. At the time of the public selection announcement, NASA will send an email to the designated small business representative indicating the outcome of the proposal package.
2. Transmission of Feedback

Phase I NASA will automatically email proposal feedback to the designated small business representative within 60 days of the announcement of selection for negotiation. If you have not received your feedback within 60 days after the announcement, contact the NASA SBIR/STTR Program Support Office at agency-sbir@mail.nasa.gov. Due to the sensitivity of this feedback, NASA will only provide feedback to the designated small business representative and not to any other parties.

Phase II Per the requirements in the email notification and this solicitation, **offerors must send a feedback request** via email to the NASA SBIR/STTR Program Office at agency-sbir@mail.nasa.gov within 60 days after the selection announcement. *Late requests will not be honored.*

5. Considerations for Contracting and Additional Information

5.1 Requirements for Negotiations

To simplify making contract awards and to reduce processing time, all contractors selected for Phase I and II contracts will ensure that:

1. All information in your proposal package is current (e.g., your address has not changed, the proposed PI is the same, etc.). If changes have occurred since submittal of your proposal package, notify the Contracting Officer immediately.
2. Your SBC is registered with System for Award Management (SAM) under the required NAICS codes (section [2.2](#)).
3. Your SBC complies with the FAR 52.222-37 Employment Reports on Special Disabled Veterans, Veterans of the Vietnam Era, and Other Eligible Veterans (VETS-4212) requirement (See the funding opportunity appendix for a list of all applicable clauses). Confirmation that a VETS-4212 report has been submitted to the Department of Labor, and is current, shall be provided to the Contracting Officer within 10 business days of the notification of selection for negotiation.
4. Your SBC HAS NOT proposed a co-principal investigator.
5. Your SBC will provide timely responses to all communications from the NSSC Contracting Officer. **Failure to respond in a timely manner to the NSSC Contracting Officer may result in your firm's selection being cancelled and no contract award.**
6. Proposed costs are supported with documentation, such as a quote, previous purchase order, published price lists, etc. All letters of commitment are dated and signed by the appropriate person with contact information. If a university is proposed as a subcontractor or a RI, the signed letter shall be on the university letterhead from the Office of Sponsored Programs. If an independent consultant is proposed, the signed letter should not be on university letterhead. If the use of Government facilities or equipment is proposed, your firm shall submit a signed letter from the Government facility authorizing the use of the facility and stating the availability and the cost, if any, together with a signed letter from your firm justifying the need to use the facility.

Failure to submit this documentation could lead to delays in the processing/negotiation of your contract and could ultimately result in proposal selection being withdrawn by NASA and no contract awarded to your firm.

Costs incurred prior to and in anticipation of award of a contract are entirely the risk of the contractor. A notification of selection for negotiation is not to be misconstrued as an award notification to commence work.

5.1.1 Requirements for Contracting

Awardees are required to make certain legal commitments through acceptance of numerous clauses in their Phase I and II contracts. This list is not a complete list of clauses to be included in Phase I and II contracts and is not the specific wording of such clauses. Copies of complete terms and conditions are available in the funding opportunity appendix.

- (1) Standards of Work. Work performed under the contract must conform to high professional standards.
- (2) Inspection. Work performed under the contract is subject to government inspection and evaluation at all times.
- (3) Examination of Records. The Comptroller General (or a duly authorized representative) must have the right to examine any pertinent records of the awardee involving transactions related to this contract.
- (4) Default. The Federal Government may terminate the contract if the contractor fails to perform the work contracted.
- (5) Termination for Convenience. The contract may be terminated at any time by the Federal Government if it deems termination to be in its best interest, in which case the awardee will be compensated for work performed and for reasonable termination costs.
- (6) Disputes. Any dispute concerning the contract that cannot be resolved by agreement must be decided by the Contracting Officer with right of appeal.

- (7) Contract Work Hours. The awardee may not require an employee to work more than 8 hours a day or 40 hours a week unless the employee is compensated accordingly (for example, overtime pay).
- (8) Equal Opportunity. The awardee will not discriminate against any employee or applicant for employment because of race, color, religion, sex, or national origin.
- (9) Equal Opportunity for Veterans. The awardee will not discriminate against any employee or application for employment because he or she is a disabled veteran or veteran of the Vietnam era.
- (10) Equal Opportunity for People with Disabilities. The awardee will not discriminate against any employee or applicant for employment because he or she is physically or intellectually disabled.
- (11) Officials Not to Benefit. No Federal Government official may benefit personally from the SBIR/STTR contract.
- (12) Covenant Against Contingent Fees. No person or agency has been employed to solicit or secure the contract upon an understanding for compensation except bona fide employees or commercial agencies maintained by the awardee for the purpose of securing business.
- (13) Gratuities. The contract may be terminated by the Federal Government if any gratuities have been offered to any representative of the government to secure the award.
- (14) Patent Infringement. The awardee must report each notice or claim of patent infringement based on the performance of the contract.
- (15) American Made Equipment and Products. When purchasing equipment or a product under the SBIR/STTR contract, purchase only American-made items whenever possible.

5.2 Awards

5.2.1 Anticipated Number of Awards

NASA does not estimate an exact number of anticipated awards; however, historical information for the program can be found in the BAA Reference Guide located in the [Firms Library](#).

5.2.2 Award Conditions

NASA awards are electronically signed by a NASA Contracting Officer and transmitted electronically to the organization via email. NSSC will distribute the NASA SBIR/STTR award with the following items:

- SF26—Contract Cover Sheet
- Contract Terms and Conditions—to include reference to the proposal package and budget
- Attachment 1: Contract Distribution List
- Attachment 2: IT Security Management Plan Template
- Attachment 3: Applicable Documents List
- Confirmation of Negotiation
- Frequently Asked Questions (FAQs)

5.2.3 Type of Contract

SBIR and STTR: NASA SBIR and STTR Phase I and II awards are firm fixed price contracts.

SBIR Ignite only: NASA SBIR Ignite Phase I award is a firm fixed price contract with a potential Phase II option.

5.2.4 Model Contracts

Examples of the NASA SBIR contracts can be found in the NASA SBIR/STTR Resources website:

https://www.nasa.gov/sbir_sttr/firms_library. *Model contracts are subject to change.*

5.3 Reporting and Required Deliverables

An IT Security Management Plan is required at the beginning of the contract. Contractors interested in doing business with NASA and/or providing IT services or solutions to NASA should use the list found at the website of the Office of the Chief Information Officer (OCIO) as a reference for information security requirements:

<https://www.nasa.gov/content/security-requirements-policies>. An example of an IT Security Management Plan can

be found in the NASA SBIR/STTR Resources website: https://www.nasa.gov/sbir_sttr/firms_library. For more information, see NASA FAR Supplement clause [1852.204-76](#).

All contracts require the delivery of technical reports that present (1) the work and results accomplished; (2) the scientific, technical, and commercial merit and feasibility of the proposed innovation and project results; (3) the proposed innovation's relevance and significance to one or more NASA interests (chapter 9); and (4) the strategy for development and transition of the proposed innovation and project results into products and services for NASA mission programs and other potential customers. Deliverables may also include the demonstration of the proposed innovation and/or the delivery of a prototype or test unit, product, or service for NASA testing and utilization if requested under Phase I.

You must provide to NASA all technical reports and other deliverables required by the contract. These reports must document progress made on the project and activities required for completion. Periodic certification for payment is required as stated in the contract. You must submit a final report to NASA upon completion of the Phase I or II R/R&D effort in accordance with applicable contract provisions.

If you receive TABA, your Phase I or II contract requires TABA deliverable(s) that summarize the outcome of the TABA services, description of the itemized technical and/or business services provided, and the actual cost for each service. TABA deliverable reports can be developed by the TABA vendor(s) with your input.

Phase I: It is standard for a TABA report to be required 30 days prior to the end of the period of performance.

Phase II: It is standard for a TABA report to be required at the end of the first year of performance and 30 days prior to the end of the period of performance.

Report deliverables must be submitted electronically. Your contract will contain instructions for submitting deliverables.

5.4 Payment Schedule

The exact payment terms are included in the contract. Invoices are submitted electronically through the Department of Treasury's Invoice Processing Platform (IPP).

If you are approved to receive TABA, the standard as to when you will submit an invoice(s) for TABA reimbursement for services rendered are at:

Phase I: 30 days prior to the end of the contract period of performance (up to \$6,500)

Phase II: The end of the first year (up to \$25,000) and 30 days prior to the end of the contract period of performance (up to \$25,000). Total Phase II TABA reimbursement is up to \$50,000.

You must submit TABA reimbursement per the payment and deliverable report schedule in your contract. NASA will not reimburse any amounts incurred over the TABA funding amount in your contract. Reimbursement must only be for actual services the TABA vendor(s) provided to the SBC during the period of performance of the contract. NASA will not reimburse any amounts of services that were not received by the SBC during the contract period of performance.

5.5 Profit

Contracts may include a reasonable profit. The reasonableness of proposed profit is determined by the Contracting Officer during contract negotiations. Reference [FAR 15.404-4](#).

5.6 Cost Sharing

Cost sharing is permitted for proposal packages under this program solicitation; however, cost sharing is not required. Cost sharing will not be an evaluation factor in consideration of your proposal package and will not be used in the determination of the percentage of Phase I work to be performed on the contract.

5.7 Rights in Data Developed Under SBIR Funding Agreements

The SBIR program provides specific rights for data developed under SBIR awards. Please review the full text at the following [FAR 52.227-20 Rights in Data-SBIR Program](#) and [PCD 21-02 FEDERAL ACQUISITION REGULATION \(FAR\) CLASS DEVIATION – PROTECTION OF DATA UNDER THE SMALL BUSINESS INNOVATIVE RESEARCH/SMALL TECHNOLOGY TRANSFER RESEARCH \(SBIR/STTR\) PROGRAM](#). If the firm wishes to exert data rights for data developed outside of this proposal, this should be proposed during contract negotiation.

5.8 Copyrights

The contractor may copyright and publish (consistent with appropriate national security considerations, if any) material developed with NASA support. NASA receives a royalty-free license for the Federal Government and requires that each publication contain an appropriate acknowledgment and disclaimer statement.

5.9 Invention Reporting, Election of Title, Patent Application Filing, and Patents

Firms shall disclose any new technology developed under the firm's contract exclusively to the National Institute of Standards and Technology (NIST) through the iEdison system per Executive Order 14014. Access to the iEdison system, including user manuals and demonstration videos, is available at <https://www.nist.gov/iedison>. Additionally, iEdison Help Desk tickets may be submitted via <https://contact.iedison.gov/>. Please review SBA SBIR/STTR Policy Directive provided in section [1.1 Legislative Authority](#) to understand these requirements.

5.10 Export Control

The contractor shall comply with all U.S. export control laws including Export Administration Regulations (EAR) and International Traffic in Arms Regulations (ITAR). Offerors are responsible for ensuring that all employees who will work on this contract⁸⁸⁸ are eligible under export control laws, EAR, and ITAR. Any employee who is not a U.S. citizen or a permanent resident may be restricted from working on this contract if the technology is restricted under export control laws, ITAR, or EAR unless the prior approval of the Department of State or the Department of Commerce is obtained via a technical assistance agreement or an export license. Violations of these regulations can result in criminal or civil penalties.

For additional information on ITAR, please visit the Code of Federal Regulations at <https://www.ecfr.gov/current/title-22/chapter-I/subchapter-M>. For additional information on EAR, please visit the Code of Federal Regulations at <https://www.ecfr.gov/current/title-15/subtitle-B/chapter-VII/subchapter-C>.

5.10 Government-Furnished and Contractor-Acquired Property

In accordance with the SBIR/STTR Policy Directive, the Federal Government may transfer title to property provided by the SBIR participating agency to the awardee or acquired by the awardee for the purpose of fulfilling the contract, where such transfer would be more cost effective than recovery of the property.

5.11 Essentially Equivalent Awards and Prior Work

Awardees must certify with every invoice that they have not previously been paid nor are currently being paid for essentially equivalent work by any agency of the Federal Government. **Failure to report essentially equivalent or duplicate efforts can lead to the termination of contracts and/or civil or criminal penalties.**

5.12 Additional Information

5.12.1 Precedence of Contract Over this Solicitation

This program solicitation reflects current planning. If there is any inconsistency between the information contained herein and the terms of any resulting SBIR contract, the terms of the contract take precedence over the solicitation.

5.12.2 Evidence of Contractor Responsibility

The Government may request you submit certain organizational, management, personnel, and financial information to establish contractor responsibility. Contractor responsibility includes all resources required for contractor performance (e.g., financial capability, workforce, and facilities).

5.13 Use of Government Resources

Federal Departments and Agencies

Use of SBIR/STTR funding for unique federal/non-NASA resources from a federal department or agency that does not meet the definition of a federal laboratory as defined by U.S. law and in the SBA Policy Directive on the SBIR/STTR program requires a waiver from the SBA. Proposal packages requiring waivers must include an explanation of why the waiver is appropriate. NASA will provide your request, along with an explanation to SBA, during the negotiation process. NASA cannot guarantee that a waiver can be obtained from SBA. Specific instructions to request use of government resources are in sections [3.1.5.4](#) of the solicitation. **NASA facilities qualify as federal laboratories.**

Support Agreements for Use of Government Resources

If you proposed the use of NASA resources, you should include a signed letter of commitment from an authorized NASA point of contact as part of your proposal submission. NASA expects selected offerors to finalize and execute their NASA SBIR Support Agreement during the negotiation period with the NSSC. All offerors selected for award who require and receive approval from the SBIR/STTR Program Executive for the use of any federal facility must, within 20 business days of notification of selection for negotiations, provide to the NSSC Contracting Officer an agreement by and between the contractor and the appropriate federal facility/laboratory, executed by the government official authorized to approve such use. The agreement must delineate the terms of use, associated costs, and facility responsibilities and liabilities. Having a signed agreement for use of government resources is a requirement for award. NASA SBIR/STTR Support Agreement template is available in the Resources website (https://www.nasa.gov/sbir_sttr/firms_library) and must be executed before a contractor can use NASA resources.

Contractor Responsibilities for Costs

In accordance with FAR Part 45, it is NASA's policy not to provide services, equipment, or facilities (resources) for the performance of work under SBIR contracts. Generally, any contractor will furnish its own resources to perform the proposed work on the contract.

In all cases, the contractor is responsible for any costs associated with services, equipment, or facilities provided by NASA or another Federal department or agency, and such costs will not increase the price of this contract.

5.14 Agency Recovery Authority and Ongoing Reporting

In accordance with 15 U.S.C. 638, NASA will –

1) require a small business concern receiving an award under its SBIR program to repay all amounts received from the Federal agency under the award if—

- (A) the small business concern makes a material misstatement that the Federal agency determines poses a risk to national security; or
- (B) there is a change in ownership, change in entity structure, or other substantial change in circumstances of the small business concern that the Federal agency determines poses a risk to national security; and

2) require a small business concern receiving an award under its SBIR program to regularly report to the Federal agency and the SBA throughout the duration of the award on—

- (A) any change to a disclosure required under subparagraphs (A) through (G) of section [2.3.1](#) above.
- (B) any material misstatement made under section [5.14](#) paragraph (A) above; and
- (C) any change described in section [5.14](#) paragraph (B) above.

6. Submission of Proposals

6.1 How to Submit Your Proposal Package

NASA uses electronically supported business processes for the SBIR/STTR program. You must have internet access and an email address. NASA will not accept paper submissions. NASA's electronic systems for submission and contract management are not document repositories. NASA follows federal regulations for records management and disposal ([NPR 1441.1E](#)).

See instructions in chapter 6 of each funding opportunity specific appendix for submissions instructions.

7. Proposal, Scientific and Technical Information Sources

7.1 NASA Organizational and Programmatic Information

General sources relating to organizational and programmatic information at NASA is available via the following websites:

NASA Budget Documents, Strategic Plans, and Performance Reports:

<http://www.nasa.gov/about/budget/index.html>

NASA Organizational Structure: <https://www.nasa.gov/organization/>

NASA SBIR/STTR Programs: https://www.nasa.gov/sbir_sttr/

Information regarding NASA's technology needs can be obtained at the following websites:

Office of Technology, Policy, and Strategy

2024 NASA Technology Taxonomy <https://www.nasa.gov/otps/2024-nasa-technology-taxonomy/>

NASA Mission Directorates

Aeronautics Research Mission Directorate (ARMD)

<http://www.aeronautics.nasa.gov/>

Exploration Systems Development Mission Directorate (ESDMD)

<https://www.nasa.gov/directorates/exploration-systems-development>

Space Operations Mission Directorate (SOMD)

<https://www.nasa.gov/directorates/space-operations-mission-directorate>

Science Mission Directorate (SMD)

<https://science.nasa.gov/>

Space Technology Mission Directorate (STMD)

<http://www.nasa.gov/directorates/spacetech/home/index.html>

NASA Centers

Ames Research Center (ARC)

<https://www.nasa.gov/ames/>

Armstrong Flight Research Center (AFRC)

<https://www.nasa.gov/armstrong/>

Glenn Research Center (GRC)

<https://www.nasa.gov/glenn/>

Goddard Space Flight Center (GSFC)

<https://www.nasa.gov/goddard/>

Jet Propulsion Laboratory (JPL)

<https://www.jpl.nasa.gov/>

Johnson Space Center (JSC)

<https://www.nasa.gov/johnson/>

Kennedy Space Center (KSC)

<https://www.nasa.gov/kennedy/>

Langley Research Center (LaRC)

<https://www.nasa.gov/langley/>

Marshall Space Flight Center (MSFC)

<https://www.nasa.gov/marshall/>

Stennis Space Center (SSC)

<https://www.nasa.gov/stennis/>

NASA Shared Services Center (NSSC)

<https://www.nasa.gov/nasa-shared-services-center/>

STMD Civil Space Shortfalls

Background and Ranking

<https://www.nasa.gov/spacetechpriorities/>

7.2 United States Small Business Administration (SBA)

The SBA oversees the Federal SBIR and STTR programs. The SBA has resources that small businesses can use to learn about the program and to get help for developing a proposal package to a Federal SBIR/STTR program. You are encouraged to review the information that is provided at the following links: www.sbir.gov, <https://www.sba.gov/local-assistance>, and at <https://www.sbir.gov/apply>.

The SBA issues a SBIR/STTR Policy Directive which provides guidance to all Federal Agencies that have a SBIR/STTR program. The Policy Directives for the SBIR/STTR programs may be obtained from the SBA at <https://www.sbir.gov/about> or at the following address:

U.S. Small Business Administration
Office of Technology – Mail Code 6470
409 Third Street, S.W.
Washington, DC 20416
Phone: 202-205-6450

7.3 National Technical Information Service

The National Technical Information Service (NTIS) is an agency of the Department of Commerce and is the Federal Government's largest central resource for government-funded scientific, technical, engineering, and business-related information. For information regarding various NTIS services and fees, email or write:

National Technical Information Service
5301 Shawnee Road
Alexandria, VA 22312
URL: <http://www.ntis.gov>
E-mail: NTRLHelpDesk@ntis.gov

8. Submission Forms

Previews of all forms and certifications are available via the NASA SBIR/STTR Resources website, located at https://www.nasa.gov/sbir_sttr/firms_library. A proposal checklist for each funding opportunity will be provided in the appendix when released.

9. Research Subtopics

Research subtopics will be released with each funding opportunity in the form of an appendix to this BAA.

Attachment A: Technology Readiness Level (TRL) Descriptions

The Technology Readiness Level (TRL) describes the stage of maturity in the development process from observation of basic principles through final product operation. The exit criteria for each level document that principles, concepts, applications, or performance have been satisfactorily demonstrated in the appropriate environment required for that level. A relevant environment is a subset of the operational environment that is expected to have a dominant impact on operational performance. Thus, reduced gravity may be only one of the operational environments in which the technology must be demonstrated or validated to advance to the next TRL.

TRL	Definition	Hardware Description	Software Description	Exit Criteria
1	Basic principles observed and reported.	Scientific knowledge generated underpinning hardware technology concepts/applications.	Scientific knowledge generated underpinning basic properties of software architecture and mathematical formulation.	Peer reviewed publication of research underlying the proposed concept/application.
2	Technology concept and/or application formulated.	Invention begins, practical application is identified but is speculative, no experimental proof or detailed analysis is available to support the conjecture.	Practical application is identified but is speculative, no experimental proof or detailed analysis is available to support the conjecture. Basic properties of algorithms, representations and concepts defined. Basic principles coded. Experiments performed with synthetic data.	Documented description of the application/concept that addresses feasibility and benefit.
3	Analytical and experimental critical function and/or characteristic proof of concept.	Analytical studies place the technology in an appropriate context and laboratory demonstrations, modeling and simulation validate analytical prediction.	Development of limited functionality to validate critical properties and predictions using non-integrated software components.	Documented analytical/experimental results validating predictions of key parameters.
4	Component and/or breadboard validation in laboratory environment.	A low fidelity system/component breadboard is built and operated to demonstrate basic functionality and critical test environments, and associated performance predictions are defined relative to the final operating environment.	Key, functionally critical, software components are integrated, and functionally validated, to establish interoperability and begin architecture development. Relevant Environments defined and performance in this environment predicted.	Documented test performance demonstrating agreement with analytical predictions. Documented definition of relevant environment.
5	Component and/or breadboard validation in relevant environment.	A medium fidelity system/component brassboard is built and operated to demonstrate overall performance in a simulated operational environment with realistic support elements that demonstrates overall performance in critical areas.	End-to-end software elements implemented and interfaced with existing systems/simulations conforming to target environment. End-to-end software system, tested in relevant environment, meeting predicted performance. Operational environment performance predicted.	Documented test performance demonstrating agreement with analytical predictions. Documented definition of scaling requirements.

		Performance predictions are made for subsequent development phases.	Prototype implementations developed.	
6	System/sub-system model or prototype demonstration in a relevant environment.	A high-fidelity system/component prototype that adequately addresses all critical scaling issues is built and operated in a relevant environment to demonstrate operations under critical environmental conditions.	Prototype implementations of the software demonstrated on full-scale realistic problems. Partially integrate with existing hardware/software systems. Limited documentation available. Engineering feasibility fully demonstrated.	Documented test performance demonstrating agreement with analytical predictions.
7	System prototype demonstration in an operational environment.	A high-fidelity engineering unit that adequately addresses all critical scaling issues is built and operated in a relevant environment to demonstrate performance in the actual operational environment and platform (ground, airborne, or space).	Prototype software exists having all key functionality available for demonstration and test. Well integrated with operational hardware/software systems demonstrating operational feasibility. Most software bugs removed. Limited documentation available.	Documented test performance demonstrating agreement with analytical predictions.
8	Actual system completed and "flight qualified" through test and demonstration.	The final product in its final configuration is successfully demonstrated through test and analysis for its intended operational environment and platform (ground, airborne, or space).	All software has been thoroughly debugged and fully integrated with all operational hardware and software systems. All user documentation, training documentation, and maintenance documentation completed. All functionality successfully demonstrated in simulated operational scenarios. Verification and Validation (V&V) completed.	Documented test performance verifying analytical predictions.
9	Actual system flight proven through successful mission operations.	The final product is successfully operated in an actual mission.	All software has been thoroughly debugged and fully integrated with all operational hardware/software systems. All documentation has been completed. Sustaining software engineering support is in place. System has been successfully operated in the operational environment.	Documented mission operational results.

Definitions

Brassboard: A medium-fidelity functional unit that typically tries to make use of as much operational hardware/software as possible and begins to address scaling issues associated with the operational system. It does not have the engineering pedigree in all aspects but is structured to be able to operate in simulated operational environments to assess performance of critical functions.

Breadboard: A low-fidelity unit that demonstrates function only, without respect to form or fit in the case of hardware, or platform in the case of software. It often uses commercial and/or ad hoc components and is not intended to provide definitive information regarding operational performance.

Engineering Unit: A high-fidelity unit that demonstrates critical aspects of the engineering processes involved in the development of the operational unit. Engineering test units are intended to closely resemble the final product (hardware/software) to the maximum extent possible and are built and tested to establish confidence that the design will function in the expected environments. In some cases, the engineering unit will become the final product, assuming proper traceability has been exercised over the components and hardware handling.

Laboratory Environment: An environment that does not address in any manner the environment to be encountered by the system, subsystem, or component (hardware or software) during its intended operation. Tests in a laboratory environment are solely for the purpose of demonstrating the underlying principles of technical performance (functions), without respect to the impact of environment.

Mission Configuration: The final architecture/system design of the product that will be used in the operational environment. If the product is a subsystem/component, then it is embedded in the actual system in the actual configuration used in operation.

Operational Environment: The environment in which the final product will be operated. In the case of spaceflight hardware/software, it is space. In the case of ground-based or airborne systems that are not directed toward spaceflight, it will be the environments defined by the scope of operations. For software, the environment will be defined by the operational platform.

Proof of Concept: Analytical and experimental demonstration of hardware/software concepts that may or may not be incorporated into subsequent development and/or operational units.

Prototype Unit: The prototype unit demonstrates form, fit, and function at a scale deemed to be representative of the final product operating in its operational environment. A subscale test article provides fidelity sufficient to permit validation of analytical models capable of predicting the behavior of full-scale systems in an operational environment.

Relevant Environment: Not all systems, subsystems, and/or components need to be operated in the operational environment to satisfactorily address performance margin requirements. Consequently, the relevant environment is the specific subset of the operational environment that is required to demonstrate critical "at risk" aspects of the final product performance in an operational environment. It is an environment that focuses specifically on "stressing" the technology advance in question.